

Fundamentals of Generative AI and Large Language Models: Theory and Practice (Intro)

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Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. I'm Sriram Ganapathy, faculty in the electrical engineering, Indian Institute of Science, Bangalore. Uh following our previous course offering on foundations of deep learning concepts and application, which received wide traction from both the career professionals and students, we are happy to launch a new course on fundamentals of generative AI and large language models. This course is catered to both UG as well as PG uh participants who may be either be students or in IT or working professionals who would like to upskill their training and skills on topics related to generative AI. The course is designed in the 12-week framework covering both introductory topics as well as practical

[00:45] hands-on sessions that happen on a weekly basis. Further, the course will have monthly live interactions with faculty participants of the course. This course is jointly offered by myself, Professor Vaishali from RV University, and Professor Ashwini from the PES University. » » I welcome uh my colleagues to talk more about the details of the course. Thank you. » Hello everyone. I'm Vaishali Gadai. Imagine asking a

computer to write a story, compose an email, generate an image, or even help you in writing code, and it does it in seconds.

[01:21] Just a few years ago, this seemed almost impossible. But today, with the rise of generative AI and large language models, it is no longer impossible. Machines are just not analyzing data, they are creating it. Across industries, companies are already using generative AI and large language models to automate customer support, accelerate software development, design new products, and gain insights from massive amount of data. These technologies are transforming how we learn, how we work, and how we innovate.

[01:58] But, an important question always remains, how these systems actually work? In this course, we will explore the foundations, the mathematical principles, the algorithms that powers these technologies. Join us as we explore the idea that enables machines to create and discover how generative AI is shaping the future of intelligent systems. » Hello everyone. I'm Ashwini Kotipalli. I'm truly delighted to be part of this NPTEL course. Let me now briefly walk you through the contents covered in this one week's course.

[02:35] In the week one, we begin with the foundations of deep learning, which is sets the base for understanding generative models. » » In week two, we move to autoencoders, one of the fundamental architectures » » in representation learning. In weeks three and four, we explore variational autoencoders or VAEs, along with advanced VAEs and their practical applications. In weeks » » five and six, we discuss generative adversarial networks, commonly known as the GANs, » » their important variants and real-world applications. In weeks seven and eight, we cover the diffusion models, which are among the most powerful modern generative techniques. In week nine, we introduce sequence models such as RNN and LSTM,

[03:33] along with the foundations of natural language processing. In week 10, we focus on the foundations of large language models and the basics of prompt engineering. In week 11, we study retrieval augmented generation or RAG, which connects large language models with external knowledge sources. » » Finally, in week 12, we look at LLM capabilities, multimodal AI, and the important ethical considerations surrounding these uh technologies. So, this course takes you step by step from core generative models all the way to modern » » large language models and their practical applications.

[04:24] » Hello all. This is Tripti. This course is designed not only to give the theoretical aspects, but also to gain the hands-on learning experience in generative AI. So, let us explore the flow of the hands-on of this particular course. » » In week 1 and 2, we'll be implementing the deep learning models such as the convolutional neural networks, apply transfer learning techniques, build ensemble models, and implement the autoencoder, which is the first generative model. In week 3 and 4, we talk about

representation learning, where we train the variational autoencoder and conditional variation autoencoder.

[05:05] In weeks 5 and 6, we talk about adversarial model development, where we implement deep convolutional generative adversarial networks, conditional GANs, and train the cycle GANs. In weeks 7 and 8, we move on to implement diffusion models by simulating the forward diffusion process and reverse diffusion training, and also develop the unit. In weeks 9 and 10, we transition into large language model implementation, wherein » » we have a hands-on on recurrent neural networks, long short-term memory, » » and transformer model development, which are the fundamentals for large language models. By the end of this journey, we will not only explore the deep learning models, but also gain experience on the generative AI

[05:57] models, which is shaping the future of artificial intelligence. Thank you. » Hello everyone. This is Debarpan. As part of the course, I'll be taking a few hands-on sessions on diffusion models and large language models. I will show how to implement the forward diffusion process and reverse denoising process in diffusion models. In large language models or LLMs, I will show how to make inferences using them, how to do prompting engineering for LLMs. Also talk about the multimodal large language models, and finally cover the problem of hallucinations in LLMs. Finally, I will conclude by showing how to use retrieval augmented generations or rag to address the problem of hallucinations.

[06:40] Thank you.